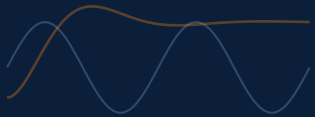
DAY 1 / 5

Mechatronics: Electrical Machines, Control, Servo Systems and Sensing



Electrical Machines and Actuators

How motors make torque, how we model them, gears and screws, motor sizing + introduction to dynamic system modelling

Mechatronics Training Programme

Theoretical Course Series ▪ Day 1 of 5

Agenda and learning objectives

Code	Session	Time
1.1	Fundamental principles and classification	0.50 h
1.2	Structure of electrical machines	1.00 h
1.3	Electrical and mechanical models	1.00 h
1.4	Actuators and motion transmission	0.75 h
1.5	Characteristics and selection criteria	0.75 h
2.1	Modelling of dynamic systems (part 1)	1.00 h
	Net teaching time	**5.00 h**

Breaks (≈10 min) roughly every hour are additional.

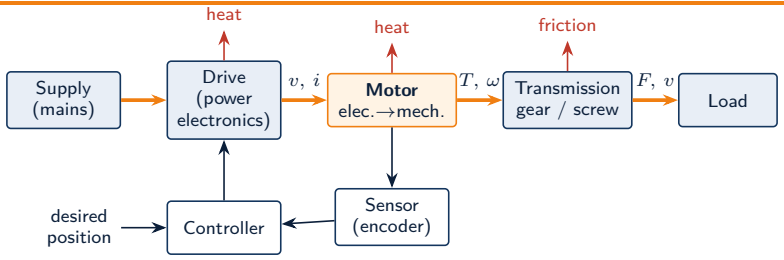
By the end of today you can . . .

- **explain** how a magnetic field turns current into torque
- **tell apart** DC, brushless, induction and stepper motors
- **write** the two DC-motor equations and **sketch** its step response
- **calculate** speed, torque and inertia through a gear or screw
- **check** whether a motor is big enough for a move
- **model** mass–spring–damper and RLC systems

Session 1

1.1 Fundamental principles and classification

A servo axis is a chain of energy converters



Refresher: power equals torque times speed

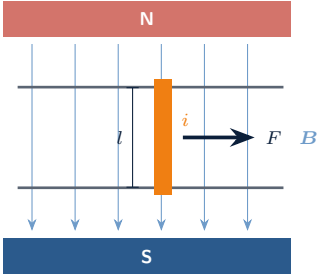
Linear: $P = F\,v$ Rotation: $P = T\,\omega$

T in N m, ω in rad/s, P in W; $\omega = \dfrac{2\pi n}{60}$ (n in rpm)

Example: 1.27 N m at 3000 rpm: $\omega = 314$ rad/s $\Rightarrow P \approx 400$ W.

- Motor: electrical power $v\,i$ in, mechanical power $T\,\omega$ out.
- Efficiency $\eta = T\omega/(v\,i) \approx 85$–$95\,\%$.
- Losses become **heat**: heat limits the motor.
- Works both ways: braking turns the motor into a **generator**.

Refresher: magnetic field and the force on a wire



wire on two rails between magnet poles

- A magnet fills space with a **magnetic field $\boldsymbol{B}$** (unit: tesla, T). Picture it as arrows from N to S.
- A wire carrying current through that field is **pushed sideways** – perpendicular to both i and $\boldsymbol{B}$.
- Direction: right-hand rule (fingers along i, bend toward $\boldsymbol{B}$, thumb = force).

$$F = B\,l\,i \qquad F \text{ [N]},\ B \text{ [T]},\ l \text{ [m]},\ i \text{ [A]}$$

Example: $B = 0.5\,\text{T}$, $l = 0.1\,\text{m}$, $i = 4\,\text{A} \Rightarrow F = 0.2\,\text{N}$.

Take-away: current + magnetic field = force. Every motor uses this.

Lorentz force and Faraday's law: the two halves of every motor

1. Motor action (Lorentz): current in a field gives force, $F = Bli$.

2. Generator action (Faraday): a wire **moving** at speed v through the field gets a voltage $e = Blv$, the **back-EMF**.

Both happen at the same time in every motor.

Bend the wire into a coil (N turns, radius r) and both laws use the same constant $k = 2NBlr$:

$$T = k_t\, i, \qquad e = k_e\, \omega, \qquad k_t = k_e \text{ (SI units)}$$

Worked example

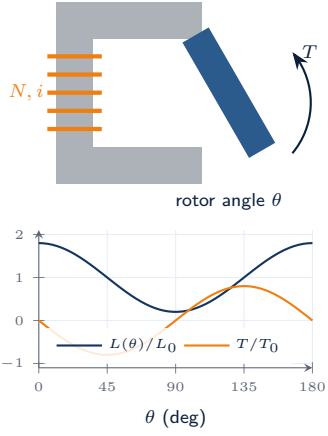
Coil: $N = 50$, $B = 0.5\,\mathrm{T}$, $l = 0.04\,\mathrm{m}$, $r = 0.02\,\mathrm{m}$

1. $k = 2 \cdot 50 \cdot 0.5 \cdot 0.04 \cdot 0.02 = 0.04\,\mathrm{N\,m/A}$
2. At $i = 5\,\mathrm{A}$: $T = 0.04 \cdot 5 = 0.2\,\mathrm{N\,m}$
3. At $\omega = 300\,\mathrm{rad/s}$: $e = 0.04 \cdot 300 = 12\,\mathrm{V}$

Take-away

Torque constant k_t (N m/A) and **back-EMF constant** k_e (V s/rad) are the same number: **current makes torque, speed makes voltage**.

Reluctance force: iron is pulled into the field



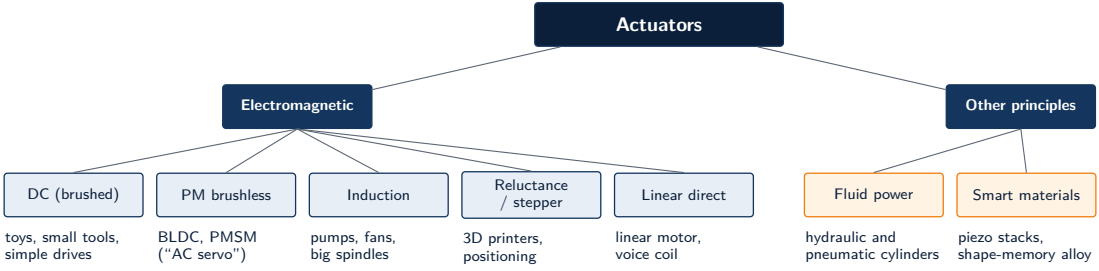
Analogy: a fridge magnet snaps onto the steel door – the iron moves to where the magnetic path is **easiest**.

- No magnet needed on the rotor: a plain iron bar turns to line up with the coil.
- Torque pushes towards the position of **highest inductance** L.
- Torque grows with i^2, so the current direction does not matter.
- Used in solenoids, **stepper motors** and switched-reluctance motors.

$$T = \tfrac{1}{2}\, i^2\, \frac{dL}{d\theta} \qquad (L \text{ in H}, \theta \text{ in rad})$$

Going further: the general co-energy method $T = \partial W'/\partial\theta$ also covers saturated iron and the extra "reluctance torque" of interior-magnet motors [Fitzgerald et al., 2003, ch. 3].

Classification of motors and actuators



	PM servo motor	Hydraulic	Pneumatic	Piezo
Force per size	high	**very high**	medium	high (tiny stroke)
Speed of response / precision	**high / high**	medium	low	**very high** / nm
Easy to control, clean?	**yes**	oil leaks, pump	air is springy	needs high voltage

Session 1.1 – Check your understanding

Check your understanding

1. A motor has $k_t = 0.1\,\mathrm{N\,m/A}$. You double the current from 2 A to 4 A. What happens to the torque?
2. A motor delivers 2 N m at 1500 rpm. Roughly how much mechanical power is that?
3. Why does a motor that should spin faster need a higher supply voltage?

How to work:

- 2 minutes alone, then compare with your neighbour.
- Write the formula first, then the numbers with units.
- Hint: $T = k_t i$, $P = T\omega$, $e = k_e \omega$.

Session 1.1 – Summary and key terms

- A motor turns $v\,i$ into $T\,\omega$ through a magnetic field – and back again when braking.
- Current in a field gives force ($F = Bli$); motion in a field gives voltage ($e = Blv$).
- Iron alone is also pulled into a field (reluctance).
- **Current limits torque, voltage limits speed, heat limits continuous use.**

Remember

$P = T\omega, \quad T = k_t\,i, \quad e = k_e\,\omega, \quad k_t = k_e$

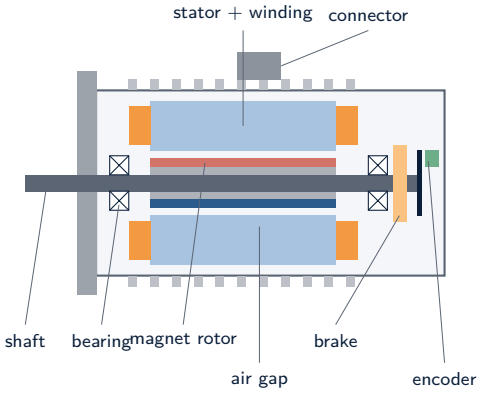
Key terms

Magnetic field B: invisible "push field" of a magnet, in tesla.
Back-EMF: voltage a spinning motor generates.
Torque constant k_t: torque per ampere.
Reluctance: "difficulty" of the magnetic path.
Efficiency η: power out / power in.

Session 2

1.2 Structure of electrical machines

Anatomy of an industrial servo motor

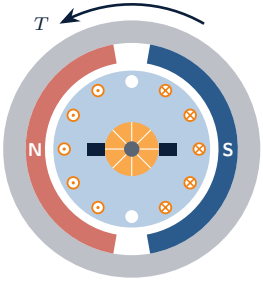


Schematic cut through a PM servo motor (not to scale).

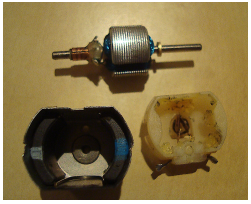
- **Stator**: the fixed outer part – iron stack with copper windings. Gets hot, so it sits next to the housing.
- **Rotor**: the turning part – shaft with magnets. Long and slim, so it is light to accelerate.
- **Air gap**: thin gap ($< 1\,\text{mm}$) where the torque is made.
- **Encoder**: measures the rotor angle (Day 5).
- Optional **brake** holds the shaft when power is off.

An industrial "AC servo motor" is usually a **PMSM**: permanent magnets on the rotor, three-phase winding on the stator, plus an encoder.

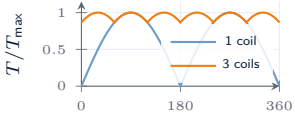
Brushed DC motor: construction and the commutator



magnets, armature, commutator



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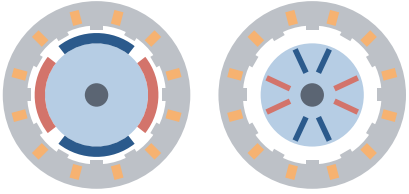
rotor angle (deg)

- **Stator magnets** make the field; the **armature** coils on the rotor carry the current.
- **Commutator + brushes**: a rotating switch that flips coil currents so torque always pushes the same way; more coils $\Rightarrow$ smoother (plot).
- Downsides: brush wear, sparks, heat trapped in the rotor.

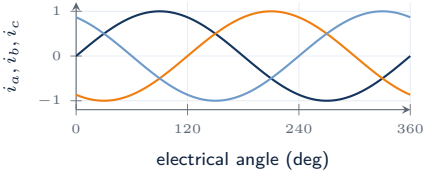
Idea for brushless motors

Replace the mechanical switch by **electronic switches + a rotor-angle sensor**.

Brushless motors (BLDC, PMSM) and electronic commutation



SPM: magnets on surface IPM: magnets buried



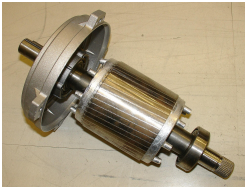
- "Inside-out" DC motor: **magnets on the rotor**, three-phase coils on the stator ⇒ no brushes, easy cooling.
- The drive switches the coil currents according to the rotor angle = **electronic commutation**.
- **BLDC**: 3 Hall sensors, 6 switching steps per turn ("block" currents) – simple, some torque ripple.
- **PMSM**: encoder gives the exact angle, currents are smooth **sine waves** 120° apart (plot) – very smooth torque.



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Going further: p pole pairs: $\omega_e = p\,\omega_m$; PMSM torque $T = \frac{3}{2}\,p\,\psi_{\mathrm{PM}}\,i_q$ via field-oriented control (Day 3–4).

Induction motor and stepper motor: short introduction



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Induction motor (squirrel-cage rotor)

- Rotating stator field **induces** currents in the rotor bars – no magnets.
- Rotor runs slightly slower than the field: **slip** s (a few %).
- Cheap, robust: pumps, fans, big spindles.

$$n_s = \frac{60f}{p}, \qquad s = \frac{n_s - n}{n_s}$$

Example: $f = 50\,\text{Hz}$, $p = 2$:
$n_s = 1500\,\text{rpm}$; at $1440\,\text{rpm}$, $s = 4\,\%$.

Stepper motor (reluctance + magnet)

- Each current pulse moves the rotor one **step**, e.g. 1.8° (200 steps/rev).
- Position by **counting steps**: no sensor needed (**open loop**).
- Risk: if the load is too heavy, steps are **lost** and nobody notices.
- Torque drops quickly with speed.

Going further: torque–slip curve (Kloss formula), vector control of induction motors, micro-stepping and stepper resonance.

Comparison of machine types for motion control

	Brushed DC	BLDC	PMSM (AC servo)	Induction	Stepper
Switching of coils	brushes	6 steps, Hall	smooth sine	induced	step pulses
Torque per size	medium	high	**very high**	medium	high at low speed
Smoothness	medium	medium	**very smooth**	smooth	jerky
Max. speed	limited	high	high	**very high**	low
Sensor needed	no	Hall	encoder	optional	none
Maintenance	brushes	none	none	none	none
Typical use	toys, tools	fans, drones	robots, machine tools	pumps, spindles	3D printers

Rule of thumb: **PMSM** for precise, dynamic axes up to ≈10 kW; induction for large, rugged, high-speed drives; stepper when cheap and slow is enough.

Session 1.2 – Check your understanding

Check your understanding

1. What job does the commutator do in a brushed DC motor – and what does that job in a BLDC motor?
2. Why is a brushless servo motor easier to cool than a brushed DC motor of the same size?
3. A stepper motor is driving a heavy load and the controller counts 200 steps. Can you be sure the shaft turned exactly one revolution? Why?

Hint: think about *where* the current flows (rotor or stator) and *who* knows the rotor angle.

Session 1.2 – Summary and key terms

- Every machine: stator, rotor, air gap.
- Coil currents must be switched as the rotor turns (**commutation**): brushes (DC), electronics + sensor (BLDC, PMSM).
- Brushless: coils on the stator ⇒ cooler, no wear.
- Induction: needs slip, no magnets. Stepper: counts steps, may lose them.

Remember

$n_s = 60f/p$, $s = (n_s - n)/n_s$, stepper: 200 steps/rev $= 1.8^\circ$

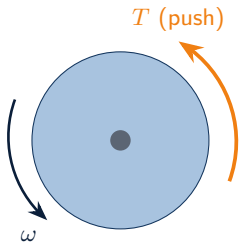
Key terms

Stator / rotor: fixed / turning part.
Commutation: switching coil currents with rotor angle.
BLDC / PMSM: brushless magnet motors, block vs. sine currents.
Slip: how much slower the rotor turns than the field.
Open loop: no measurement of the result.

Session 3

1.3 Electrical and mechanical models

Refresher: Newton's law for rotation



disc with inertia J

Linear	Rotation	Unit
force F	torque T	N m
mass m	inertia J	kg m^2
velocity v	angular speed ω	rad/s
$F = m\,\dfrac{dv}{dt}$	$T = J\,\dfrac{d\omega}{dt}$	

Analogy: a heavy playground roundabout is hard to start and hard to stop – it has a large **moment of inertia** J.

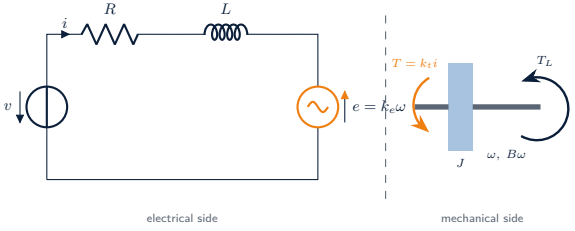
$$J\,\frac{d\omega}{dt} = \sum T = T_{\text{motor}} - T_{\text{friction}} - T_{\text{load}}$$

Worked example

1. Rotor $J = 1 \times 10^{-4}$ kg m^2, net torque $T = 0.5$ N m.
2. Acceleration $\alpha = T/J = 5000$ rad/s^2.
3. Time to reach 314 rad/s (3000 rpm): $t = 314/5000 = 0.063$ s.

Disc: $J = \frac{1}{2}mr^2$. **Take-away:** net torque sets how fast the speed changes.

Equivalent circuit of the DC motor



The electrical side is just the **RL circuit** you know, plus one extra "battery" $e = k_e\omega$ that grows with speed and **opposes** the supply.

Sym.	Meaning	Unit
R	winding resistance	Ω
L	winding inductance	H
k_e	back-EMF constant	V s/rad
k_t	torque constant	N m/A
J	rotor (+ load) inertia	kg m^2
B	viscous friction	N m s/rad
T_L	load torque	N m

Signs: v, i positive into the motor; T_L works against the motion.

The two motor equations

Electrical (KVL around the loop):

$$v = R\,i + L\frac{di}{dt} + k_e\,\omega$$

Mechanical (Newton for rotation):

$$J\frac{d\omega}{dt} = k_t\,i - B\,\omega - T_L$$

- Linked both ways: i makes torque, ω makes back-EMF.
- Two first-order ODEs $\Rightarrow$ one **second-order** system.

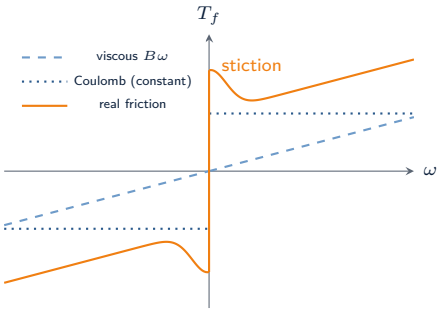
Why k_t equals k_e: energy conservation

Power converted on the electrical side: $e\,i = k_e\,\omega\,i$
Power delivered on the mechanical side: $T\,\omega = k_t\,i\,\omega$
They must be equal $\Rightarrow$ $\boldsymbol{k_t = k_e}$ (SI units).

Datasheet units: k_e is often given in V per 1000 rpm.
Divide by 104.7 to get V s/rad:
10.47 V/krpm $\Rightarrow$ $k_e = k_t = 0.1$ SI.

Going further: three-phase datasheets use rms and line-to-line values, giving $k_t = \sqrt{3}\,k_e$; resistance rises 0.4 %/K when hot.

Friction and load torque



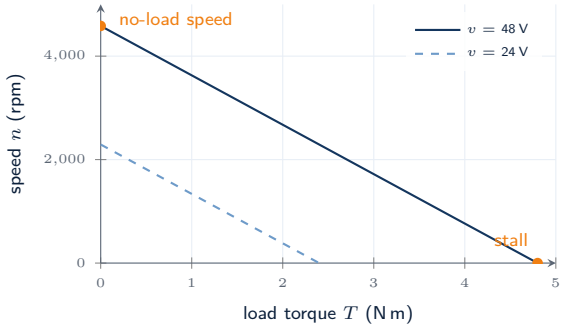
Analogy: pushing a heavy box – hard to get it moving, easier once it slides.

- **Viscous** friction $B\omega$: grows with speed, like stirring honey. Linear $\Rightarrow$ easy maths.
- **Coulomb** friction: constant, flips sign with direction.
- **Stiction**: extra "sticking" torque at standstill $\Rightarrow$ jerky start, small position errors.
- **Load torque** T_L: what the process demands – gravity on a vertical axis, cutting forces, springs.

model used today: $T_f \approx B\,\omega$ (Day 2: the rest)

[Armstrong-Hélouvry, Dupont & Canudas de Wit, 1994]

Steady state: the speed–torque line



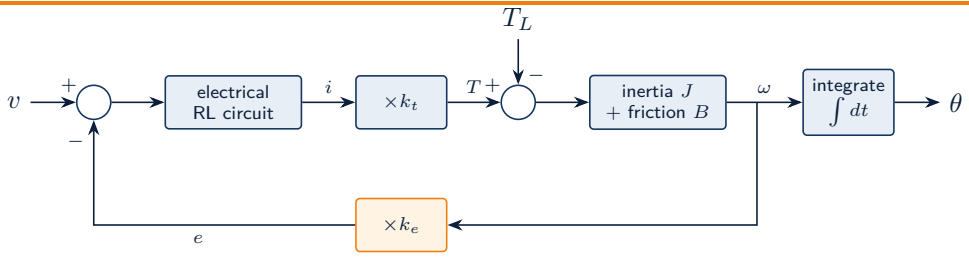
Example motor: $R = 1\,\Omega$, $k_t = k_e = 0.1$ SI, friction neglected.

Constant speed: set $\frac{d}{dt} = 0$ in both equations:

$$\omega = \frac{v}{k_e} - \frac{R}{k_e k_t} T$$

1. No load: $\omega_0 = 48/0.1 = 480$ rad/s (4584 rpm).
2. Stalled: $i = v/R = 48$ A, $T = k_t i = 4.8$ N m.
3. More load $\Rightarrow$ speed **droops** along the line.
4. Higher voltage $\Rightarrow$ line moves up.

Block diagram of the DC motor

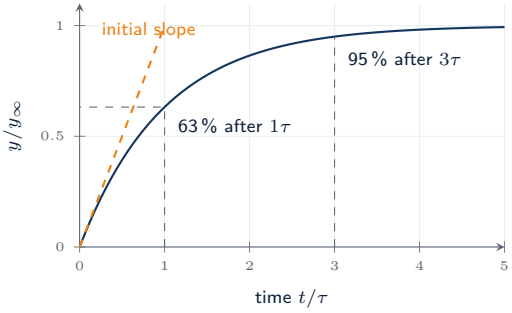


- A **block diagram** draws the equations as signal flow: boxes act on signals, circles add or subtract.
- Back-EMF is a built-in **feedback loop**: faster ⇒ more e ⇒ less current ⇒ less torque. The motor calms itself.
- T_L enters as a **disturbance**.

> Real servo drives control the **current** directly, so $T \approx k_t\, i^*$: the motor behaves like a **torque source** (Day 4).

Going further: with the Laplace transform each box becomes a transfer function, e.g. $1/(Ls + R)$ and $1/(Js + B)$ (Day 2).

Refresher: first-order ODEs and the exponential response



You know this from RC charging: the capacitor voltage rises fast at first, then slower and slower.

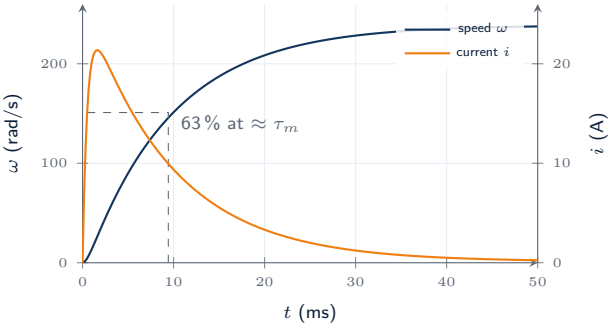
$$\tau \frac{dy}{dt} + y = y_\infty \;\Rightarrow\; y(t) = y_\infty\left(1 - e^{-t/\tau}\right)$$

- τ = **time constant** [s]: RC circuit $\tau = RC$, RL circuit $\tau = L/R$.
- After 1τ: 63 %; after 3τ: 95 %; after 5τ: "done".

Example: $L = 0.5\,\text{mH}$, $R = 1\,\Omega \Rightarrow \tau = 0.5\,\text{ms}$; current settles in $\approx 1.5\,\text{ms}$.

Take-away: one energy store $\Rightarrow$ one τ $\Rightarrow$ one exponential.

The motor's two time constants: worked example



Simulated 24 V step, $R = 1\,\Omega$, $L = 0.5$ mH, $k_t = k_e = 0.1$, $J = 1 \times 10^{-4}$ kg m^2.

1. Electrical: $\tau_e = L/R = 0.5$ ms.
2. Mechanical: $\tau_m = \dfrac{RJ}{k_t k_e} = \dfrac{1 \cdot 10^{-4}}{0.01} = 10$ ms.
3. Final speed: $\omega_\infty \approx v/k_e = 240$ rad/s.
4. Start current: $\omega = 0$, no back-EMF $\Rightarrow$ up to $v/R = 24$ A (peak 21 A).

$$\tau_e = \frac{L}{R} \quad \ll \quad \tau_m = \frac{RJ}{k_t k_e}$$

Take-away: current is fast, speed is slow; more load inertia $\Rightarrow$ larger τ_m. Real drives limit the start current.

Session 1.3 – Check your understanding

Check your understanding

1. A motor has $\tau_m = 20\,\text{ms}$. Roughly how long until it reaches 95 % of its final speed after a voltage step?
2. You attach a load that doubles the total inertia J. What happens to τ_m?
3. Why is the current so large in the first milliseconds after switching on, and why does it then fall?

Hint: use the 63 % / 95 % rule, the formula $\tau_m = RJ/(k_t k_e)$ and the back-EMF $e = k_e \omega$.

Session 1.3 – Summary and key terms

- Motor = RL circuit + back-EMF, coupled to an inertia with friction and load.
- Energy conservation gives $k_t = k_e$ (SI units).
- Constant speed: straight speed–torque line.
- Two time constants: fast electrical τ_e, slow mechanical τ_m.

Remember

$$v = Ri + L\frac{di}{dt} + k_e\omega, \quad J\frac{d\omega}{dt} = k_t i - B\omega - T_L$$
$$\tau_e = L/R, \quad \tau_m = RJ/(k_t k_e)$$

Key terms

Moment of inertia J: resistance to changes of rotation speed.
Time constant τ: time to reach 63 %.
Stiction: sticking friction at standstill.
Block diagram: picture of equations as signal flow.
Disturbance: unwanted input such as T_L.

Session 4

1.4 Actuators and motion transmission

Motion transmission options

Mechanism	Converts	Speed	Stiffness	Play (backlash)
Ball screw	rotation → linear	$\leq$ 1–2 m/s	high	very small
Toothed belt	rotation → linear	$\leq$ 5–10 m/s	low	small
Rack & pinion	rotation → linear	2–5 m/s	medium	some
Gearbox	fast rotation → slow	motor-limited	medium	small–some
Linear motor / direct drive	none needed	high	highest	none

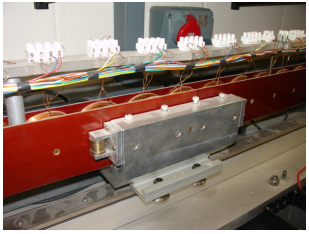
With gear / screw

+ small, fast, cheap motor; load "looks lighter"
− play, springiness, friction, wear, noise

Direct drive

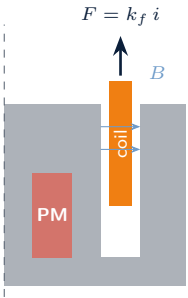
+ stiff, no play, very precise
− big, expensive motor; feels the full load

Linear actuators: linear motor and voice coil



Zureks, CC BY-SA 4.0, via Wikimedia Commons

- **Linear motor** = a PMSM "unrolled" flat: coil carriage + magnet track.
- Very fast and precise, no screw needed.



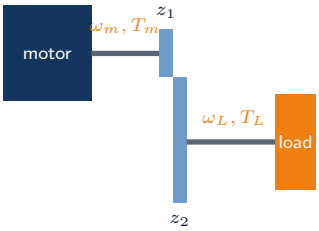
voice coil, half-section

Voice coil: like a loudspeaker – a coil in a magnet gap moves in and out.

$$F = k_f\, i, \quad m\frac{dv}{dt} = k_f\, i - b\, v - F_L$$

- $k_f = NBl$ [N/A] – the force law from Session 1.1.
- Short stroke (mm–cm), extremely fast: hard-disk heads, camera focus.
- Same equations as the DC motor with $T \to F$, $\omega \to v$, $J \to m$.

Refresher: gear ratio – think bicycle gears



small gear z_1 teeth drives big gear z_2

Bicycle: low gear uphill – you pedal fast and easily, the wheel turns slowly with a big push.

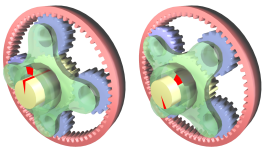
$$N = \frac{z_2}{z_1} = \frac{\omega_m}{\omega_L}, \qquad T_L = N\,T_m \text{ (ideal)}$$

- Speed goes **down** by N, torque goes **up** by N.
- Power is unchanged (no losses): $T_L\omega_L = T_m\omega_m$.

Worked example: 20 teeth drive 60 teeth

1. $N = 60/20 = 3$.
2. Motor 3000 rpm $\Rightarrow$ load $3000/3 = 1000$ rpm.
3. Motor 1 N m $\Rightarrow$ load $3 \cdot 1 = 3$ N m.
4. Power: $1 \cdot 314 = 3 \cdot 105 \approx 314$ W on both sides.

Gearboxes: planetary and harmonic drives



Wapcaplet, CC BY-SA 3.0, via Wikimedia Commons



Pieceofmetalwork, CC BY-SA 4.0, via Wikimedia Commons

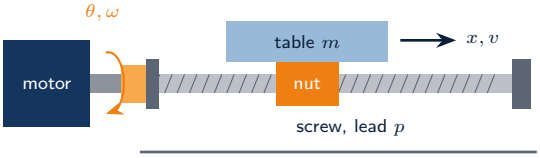
- **Planetary gear**: sun gear in the middle, planets around it, ring outside. Compact, efficient, ratio 3–10 per stage.
- **Harmonic drive**: a flexible toothed cup is squeezed into an oval – ratio 50–160 in one stage, almost no play. Used in robot joints.

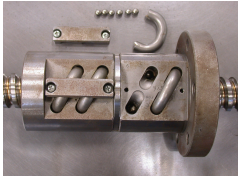
planetary (ring fixed): $N = 1 + \dfrac{z_{\text{ring}}}{z_{\text{sun}}}$

Example: $z_{\text{ring}} = 60$, $z_{\text{sun}} = 20 \Rightarrow N = 4$.

	Ratio / stage	Efficiency	Play	Typical use
Planetary	3–10	95–98 %	small	servo axes
Harmonic drive	50–160	60–90 %	≈ 0	robot joints
Worm	5–100	40–90 %	medium	self-locking lifts

Ball screw: from rotation to straight-line motion





Glenn McKechnie (Graibeard), CC BY-SA 3.0, via Wikimedia Commons

Analogy: a nut on a bolt – turn the bolt once, the nut moves by one thread pitch. The **lead** p is that distance per turn.

$$v = \frac{p}{2\pi}\,\omega \qquad (p \text{ in m/rev})$$

Worked example: lead 10 mm

1. $3000\,\text{rpm} = 50\,\text{rev/s}$.
2. $v = 50 \cdot 0.01\,\text{m} = 0.5\,\text{m/s}$.

Balls roll between screw and nut $\Rightarrow \eta \approx 0.9$, very little play.

Reflected inertia: what the motor “feels”



equivalent load at the motor shaft

Bicycle again: in a low gear the bike feels **light** to accelerate – but you must pedal fast.

Kinetic energy must be the same on both sides:

$\frac{1}{2} J_L \omega_L^2 = \frac{1}{2} \frac{J_L}{N^2} \omega_m^2 \quad (\text{since } \omega_L = \omega_m / N)$

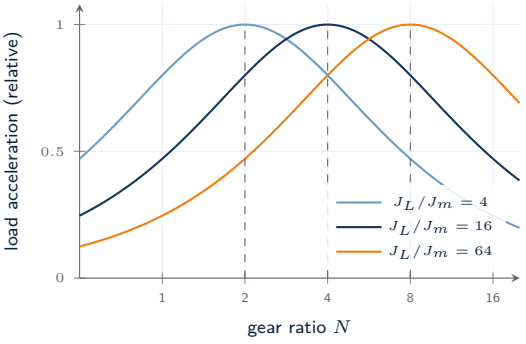
$$J_{\text{tot}} = J_m + \frac{J_L}{N^2}, \qquad \text{load torque seen: } \frac{T_L}{N}$$

Why N^2? The load turns $N\times$ slower (one factor) *and* the gear multiplies torque by N (second factor).

Example: $J_L = 9 \times 10^{-4}\,\text{kg m}^2$, $N = 3 \Rightarrow$ $J_L/N^2 = 1 \times 10^{-4}\,\text{kg m}^2$.

Mechanism	**Inertia at motor**
Gear, ratio N	J_L/N^2
Ball screw, lead p	$m\,(p/2\pi)^2$
Belt / rack, radius r	$m\,r^2$

Choosing the gear ratio: inertia matching



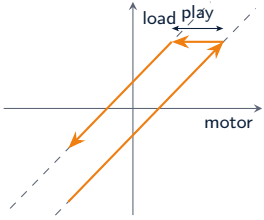
- N too small: motor must fight the full load inertia.
- N too large: motor spins very fast and wastes torque on its **own** inertia.
- Best acceleration in between (peak of each curve):

$$N_{\text{opt}} = \sqrt{J_L/J_m} \;\Leftrightarrow\; J_L/N^2 = J_m$$

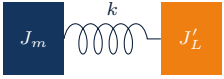
Example: $J_L/J_m = 16 \Rightarrow N_{\text{opt}} = 4$.

- In practice the motor's **maximum speed** often limits N.
- Rule: keep **inertia ratio** $J_L/(N^2 J_m)$ below $\approx$ 3–10.

Real transmissions: losses, play and springiness



backlash: on reversal the motor turns, the load does not



a soft belt or shaft acts like a **spring** between motor and load $\Rightarrow$ the load can wobble (**resonance**, Session 2.1)

- **Efficiency** $\eta < 1$: part of the torque is lost to friction ($T_{\text{out}} = \eta N T_{\text{in}}$).
- **Backlash**: lost motion when direction changes – position errors.
- **Stiffness** k: belts and harmonic drives twist under load.
- Fixes: preloaded nuts, stiff couplings, load-side encoder (Day 5).

Session 1.4 – Check your understanding

Check your understanding

1. A gearbox has $N = 5$. The motor runs at 2500 rpm with 0.4 N m. What are the load speed and (ideal) load torque?
2. A load inertia of 0.05 kg m^2 sits behind that $N = 5$ gearbox. What inertia does the motor feel?
3. Why does backlash cause trouble exactly when the axis changes direction?

Hint: speed $\div N$, torque $\times N$, inertia $\div N^2$.

Session 1.4 – Summary and key terms

- Gears and screws trade speed for torque/force.
- The load is "reflected" to the motor shaft by **energy equivalence**: inertia $\div N^2$.
- Best acceleration: $N_{\text{opt}} = \sqrt{J_L/J_m}$; speed limits often decide.
- Linear motors and voice coils: same equations with F, v, m.

Remember

$$N = \omega_m/\omega_L, \quad J'_L = J_L/N^2, \quad v = \frac{p}{2\pi}\omega, \quad J'_{\text{screw}} = m\left(\frac{p}{2\pi}\right)^2$$

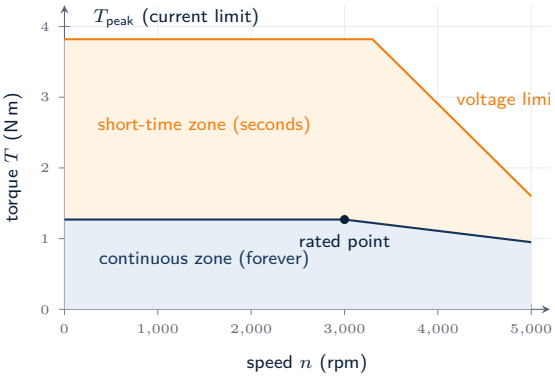
Key terms

Gear ratio N: motor speed / load speed.
Lead p: travel per screw revolution.
Reflected inertia: load inertia as felt by the motor.
Backlash: play in a transmission.
Stiffness: resistance to twisting/stretching.

Session 5

1.5 Characteristics and selection criteria

Servo motor torque–speed envelope

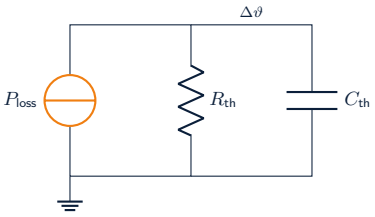


Typical 400 W AC servo: $T_N = 1.27$ N m, $T_{\text{peak}} = 3.82$ N m, $n_N = 3000$ rpm.

Analogy: you can sprint for 10 seconds, but you can only jog for an hour.

- **Continuous torque**: the motor can give this forever without overheating.
- **Peak torque**: about $3\times$ more, but only for short bursts (acceleration).
- **Voltage limit**: at high speed the back-EMF "eats" the supply voltage, so less current – and torque – gets in.

Thermal limits: the motor as a first-order system

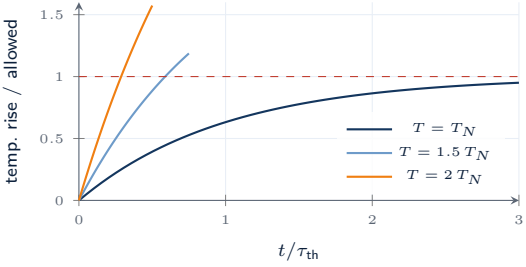


heat flows like current

$$\Delta\vartheta(t) = P_{\text{loss}} R_{\text{th}} \left(1 - e^{-t/\tau_{\text{th}}}\right)$$

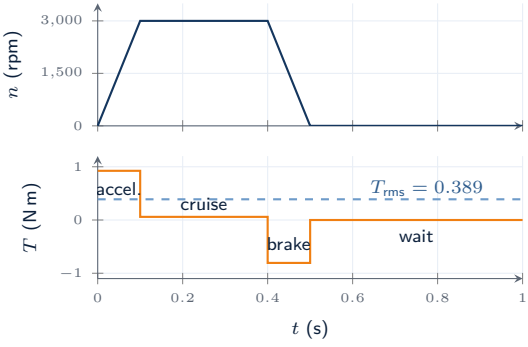
Same exponential as the RC circuit!

$\tau_{\text{th}} = R_{\text{th}} C_{\text{th}}$.



- $\Delta\vartheta$: temperature rise [K]; R_{th}: thermal resistance [K/W]; C_{th}: heat capacity [J/K].
- Copper loss $\propto i^2 \propto T^2$: $2\times$ torque $= 4\times$ heat.
- Example: $P = 50\,\text{W}$, $R_{\text{th}} = 1.6\,\text{K/W} \Rightarrow$ final rise $80\,\text{K}$.
- τ_{th} is minutes $\Rightarrow$ short overloads are fine.

Duty cycle and RMS torque



Heat $\propto T^2$, and the motor heats up slowly, so it "feels" the **average of T^2** over the cycle:

$$T_{\text{rms}} = \sqrt{\frac{1}{t_{\text{cyc}}} \sum_k T_k^2 \, t_k} \;\le\; T_{\text{cont}}$$

- T_k: torque in segment k, lasting t_k seconds.
- Why not the plain average? Braking torque is negative but heats just as much.
- Also check: largest $|T_k| \le T_{\text{peak}}$, max speed $\le n_{\text{max}}$.
- Vertical axis: holding against gravity during "wait" also counts!

Worked sizing example (1/2): move and inertia

Given: horizontal ball-screw axis

Table + part $m = 50\,\text{kg}$
Screw lead $p = 10\,\text{mm}$; screw inertia $J_s = 1.0 \times 10^{-4}\,\text{kg m}^2$
Friction torque (all) $T_f = 0.059\,\text{N m}$
Move: accelerate 0.1 s, cruise 0.3 s, brake 0.1 s, wait 0.5 s
Top speed $0.5\,\text{m/s}$

Step 1 – motor speed
$n = v/p = 0.5/0.01 = 50\,\text{rev/s} = 3000\,\text{rpm} \Rightarrow \omega = 314\,\text{rad/s}$

Step 2 – acceleration
$\alpha = \omega/t_a = 314/0.1 = 3140\,\text{rad/s}^2$

Step 3 – table inertia seen by the motor
$J_t = m\,(p/2\pi)^2 = 50 \cdot (0.01/6.28)^2 = 1.27 \times 10^{-4}\,\text{kg m}^2$

Step 4 – total load inertia (incl. coupling 0.1×10^{-4})
$J_L = 1.0 + 1.27 + 0.1 = 2.37 \times 10^{-4}\,\text{kg m}^2$

> Screw inertia grows with diameter^4: $J_s = \pi\rho L d^4/32$ (here $d = 20\,\text{mm}$, $L = 0.8\,\text{m}$, steel).

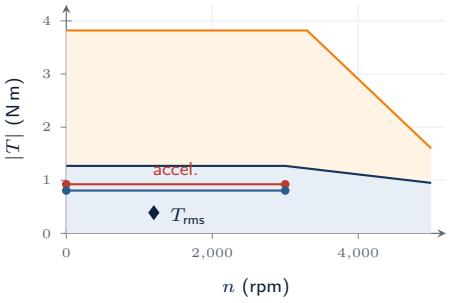
Worked sizing example (2/2): torques and checks

Step 5 – candidate: 400 W servo, $J_m = 0.40 \times 10^{-4}$ kg m^2, $T_N = 1.27$ N m, $T_{\text{peak}} = 3.82$ N m. $J_{\text{tot}} = 2.37 + 0.40 = 2.77 \times 10^{-4}$ kg m^2 (ratio $J_L/J_m \approx 6$ ✓)

Step 6 – torque per segment ($T = J\alpha \pm T_f$)
accel. $0.87 + 0.06 = 0.92$ N m; cruise 0.06 N m; brake $-0.87 + 0.06 = -0.81$ N m; wait 0

Step 7 – RMS torque
$T_{\text{rms}} = \sqrt{(0.92^2 \cdot 0.1 + 0.06^2 \cdot 0.3 + 0.81^2 \cdot 0.1)/1} =$ **0.39 N m**

Step 8 – checks
peak $0.92 < 3.82$ ✓ RMS $0.39 < 1.27$ ✓ speed $3000 \leq 3000$ ✓



All points inside the envelope: the motor is big enough, with 3× thermal margin.

Motor–drive matching and selection checklist

Voltage: at top speed the drive must beat the back-EMF:

$$V_{\text{supply}} \gtrsim k_e\,\omega_{\text{max}} + R\,I_{\text{peak}}$$

Current: $I_{\text{peak}} = T_{\text{peak}}/k_t$ must be within the drive's rating.

Example ($k_e = k_t = 0.3$ SI):

1. Back-EMF at 314 rad/s: $0.3 \cdot 314 \approx 94$ V.
2. A 48 V drive is too small; a 230 V drive (325 V DC) is fine.
3. $I_{\text{peak}} = 0.92/0.3 \approx 3$ A.

Going further: rms/line-to-line datasheet conventions, inductive voltage drop, braking (regenerated) energy.

Selection checklist

1. Describe the move, the load and the cycle.
2. Reflect inertia and torques to the motor.
3. Choose the gear/screw ratio.
4. Peak torque inside the **short-time zone**.
5. T_{rms} inside the **continuous zone** (+20–30 % margin).
6. Inertia ratio OK for the mechanics.
7. Drive: voltage and current.
8. Encoder, brake, protection class.

Session 1.5 – Check your understanding

Check your understanding

1. A motor can deliver $3\times$ its rated torque. Why can it not do so permanently?
2. In a 1-second cycle the motor gives 1 N m for 0.1 s and zero the rest of the time. What is T_{rms}?
3. Why do we use RMS torque instead of the ordinary average torque for the thermal check?

Hint: heat $\propto T^2$; the motor's thermal time constant is minutes.

Session 1.5 – Summary and key terms

- The torque–speed envelope has a continuous (thermal) zone and a short-time (current/voltage) zone.
- The motor warms up like an RC circuit charges: exponentially, over minutes.
- Judge heating with T_{rms}, acceleration with T_{peak}.
- Size step by step: speed $\rightarrow$ inertia $\rightarrow$ torques $\rightarrow$ RMS/peak $\rightarrow$ drive.

Remember

$$T_{\text{rms}} = \sqrt{\frac{1}{t_{\text{cyc}}} \sum T_k^2 t_k}, \qquad V \gtrsim k_e \omega_{\text{max}} + R I_{\text{pk}}$$

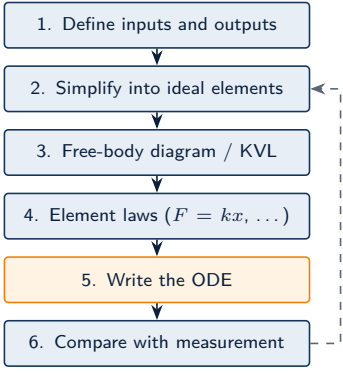
Key terms

Continuous torque: torque possible forever.
Peak torque: short-burst torque.
RMS torque: "heating-equivalent" torque of a cycle.
Duty cycle: repeating pattern of the move.
Thermal resistance: K of rise per W of loss.

Session 6

2.1 Modelling of dynamic systems – fundamental physical models

Lumped-parameter modelling: workflow and building blocks

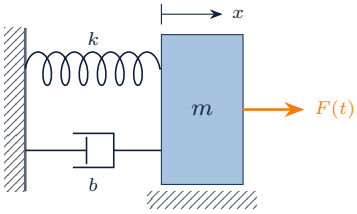


Domain	Motion store	Position store	Loss
Translation	mass m	spring k	damper b
Rotation	inertia J	torsion spring k_θ	damper B
Electrical	inductor L	capacitor C	resistor R

- **Lumped**: each property sits in one ideal element (a mass is a point, a spring has no mass).
- Each energy store adds one first-order equation $\Rightarrow$ the **order** of the system.
- Use the **simplest model that answers your question**.

[Karnopp, Margolis & Rosenberg, 2012; Ogata, 2010]

Mass–spring–damper: from free-body diagram to ODE



free-body diagram

Analogy: a car's suspension – spring carries the car, shock absorber (damper) stops the bouncing.

Newton $\sum F = m\ddot{x}$ with spring force kx and damper force $b\dot{x}$:

$$m\,\ddot{x} + b\,\dot{x} + k\,x = F(t)$$

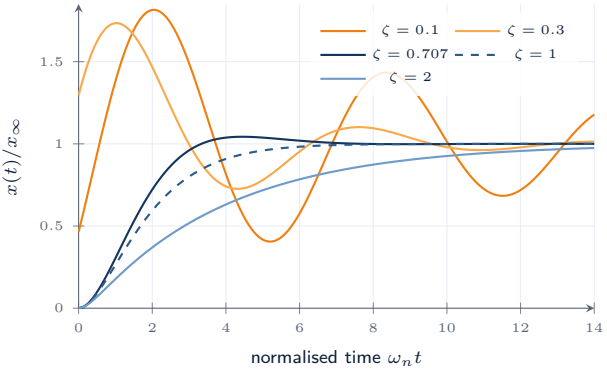
Two numbers describe it

$\omega_n = \sqrt{k/m}$ **natural frequency** [rad/s]: how fast it bounces

$\zeta = \dfrac{b}{2\sqrt{km}}$ **damping ratio** [–]: how quickly bouncing dies

Example: $m = 2\,\text{kg}$, $k = 2 \times 10^5\,\text{N/m}$, $b = 120\,\text{N s/m}$:
$\omega_n = \sqrt{10^5} = 316\,\text{rad/s}$ (50 Hz),
$\zeta = 120/1265 = 0.095$.

Second-order step responses: the role of damping



Push the mass with a constant force (a **step**) and watch $x(t)$:

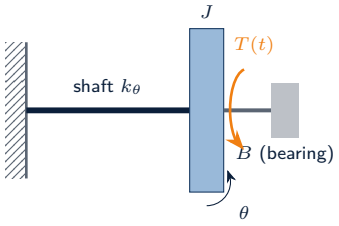
- $\zeta < 1$: **overshoots and rings** – a car with worn shock absorbers.
- $\zeta \approx 0.7$: fast, tiny overshoot (4 %) – a good design target.
- $\zeta \geq 1$: no overshoot, but sluggish.

overshoot $\approx e^{-\pi\zeta/\sqrt{1-\zeta^2}}$

$\zeta = 0.1$: 73 %, $\zeta = 0.3$: 37 %.

Going further: the poles $s = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$ in the complex plane (Day 2).

Rotational mechanical systems

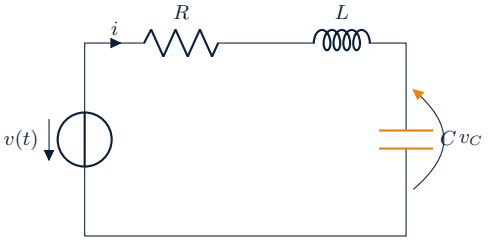


$$J\,\ddot{\theta} + B\,\dot{\theta} + k_\theta\,\theta = T(t)$$

Exactly the mass–spring–damper with $m \to J$, $b \to B$, $k \to k_\theta$, $F \to T$.

- Newton for rotation: $\sum T = J\ddot{\theta}$.
- A shaft twists like a spring: torsional stiffness k_θ [N m/rad].
- Solid shaft: $k_\theta = \dfrac{G\,\pi d^4}{32\,l}$ (G: shear modulus).
- Example: steel ($G = 80$ GPa), $d = 10$ mm, $l = 100$ mm $\Rightarrow k_\theta \approx 785$ N m/rad.
- Couplings, belts and gear teeth are springs too – the **softest one dominates**.
- Motor without load: $k_\theta = 0$, so only $J\ddot{\theta} + B\dot{\theta} = T$.

Electrical RLC network



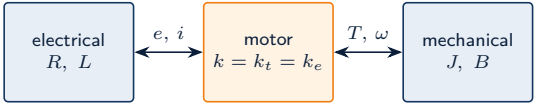
$$L\,\ddot{q} + R\,\dot{q} + \frac{1}{C}\,q = v(t), \qquad i = \dot{q}$$

- KVL: $v = v_R + v_L + v_C$ with $v_R = Ri$, $v_L = L\,di/dt$, $v_C = q/C$.
- Write it with the charge q: the **same form** as $m\ddot{x} + b\dot{x} + kx = F$!
- L acts like mass, R like damper, $1/C$ like spring.
- $\omega_n = \dfrac{1}{\sqrt{LC}}, \quad \zeta = \dfrac{R}{2}\sqrt{\dfrac{C}{L}}.$
- Example: $L = 1\,\text{mH}$, $C = 470\,\mu\text{F}$, $R = 0.1\,\Omega \Rightarrow$ $f_n = 232\,\text{Hz}$, $\zeta = 0.034$: rings a lot.

Analogies: one equation, three domains – and the motor links them

Translation	Rotation	Electrical
force F	torque T	voltage v
velocity $\dot{x}$	speed ω	current i
mass m	inertia J	inductance L
damper b	damper B	resistance R
spring k	torsion k_θ	$1/C$
$\frac{1}{2}m\dot{x}^2$	$\frac{1}{2}J\omega^2$	$\frac{1}{2}Li^2$

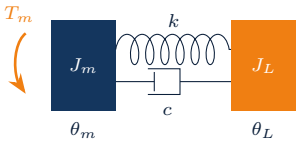
Always: (push) $\times$ (flow) $=$ **power**: Fv, $T\omega$, vi.



- The motor is the bridge: it converts power between the domains without storing any.
- Two energy stores (L and J) $\Rightarrow$ 2nd-order system, as in Session 1.3.

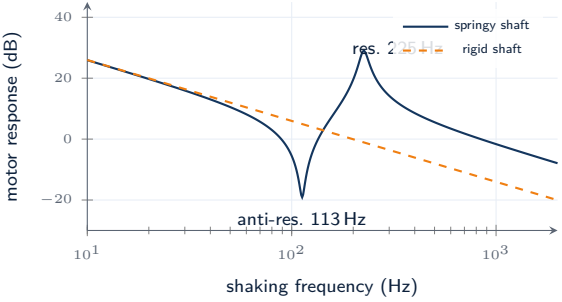
Going further: energy-based modelling with Lagrange's equations; "gyrator" in bond-graph language; state-space form $\dot{\boldsymbol{x}} = A\boldsymbol{x} + B\boldsymbol{u}$ (Day 2).

Preview: two-inertia system (motor – springy shaft – load)



Analogy: a playground swing – push at just the right rhythm and it swings higher and higher: **resonance**.

$$\omega_r = \sqrt{\frac{k\,(J_m + J_L)}{J_m\,J_L}}$$



$J_m = 2 \times 10^{-4}$, $J_L = 6 \times 10^{-4}\,\mathrm{kg\,m^2}$, $k = 300\,\mathrm{N\,m/rad}$

- Plot: how strongly the motor moves when driven by a sine torque of each frequency.
- The resonance peak limits how fast the servo can be tuned (Day 3, Day 5).

Session 2.1 – Check your understanding

Check your understanding

1. You fit a stiffer spring ($4 \times k$) to a mass–spring–damper. What happens to the natural frequency ω_n?
2. In the RLC circuit, which element plays the role of the mass? Which plays the damper?
3. A step response overshoots by 40 %. Should you increase or decrease the damping b? Why?

Hint: $\omega_n = \sqrt{k/m}$, $\zeta = b/(2\sqrt{km})$, and the analogy table.

Session 2.1 (part 1) – Summary and key terms

- Lumped models: masses, springs, dampers (and L, C, R).
- Mass–spring–damper, shaft and RLC circuit share one 2nd-order ODE with ω_n and ζ.
- Low damping $\Rightarrow$ overshoot and ringing.
- The motor links the electrical and mechanical worlds.
- A springy shaft between motor and load creates resonance.

Remember

$$m\ddot{x} + b\dot{x} + kx = F, \quad \omega_n = \sqrt{k/m}, \quad \zeta = \frac{b}{2\sqrt{km}}$$

Key terms

Lumped model: system built from ideal elements.
Natural frequency ω_n: speed of free oscillation.
Damping ratio ζ: how fast oscillation dies.
Step response: reaction to a sudden constant input.
Resonance: big response at one special frequency.

Day 1 summary

Machines

Current + field = force; motion + field = voltage; $k_t = k_e$.
DC (brushes), BLDC/PMSM (electronic), induction (slip), stepper (steps).

Models

$v = Ri + L\dot{i} + k_e\omega, \quad J\dot{\omega} = k_t i - B\omega - T_L$
Fast $\tau_e = L/R$, slow $\tau_m = RJ/(k_t k_e)$.

Transmissions and sizing

Speed $\div N$, torque $\times N$, inertia $\div N^2$; $N_{\text{opt}} = \sqrt{J_L/J_m}$.
Peak torque in the short-time zone, T_{rms} in the continuous zone.

Dynamic modelling

Lumped elements, analogies; 2nd-order system with ω_n, ζ; resonance.

One sentence

A servo axis is an **energy converter** (motor) plus a **mechanical network** (transmission, load) – both described by the same kind of ODEs.

Preview of Day 2: System dynamics and control

Code	Session	Time
2.1	Modelling (cont.): nonlinear effects and linearisation	0.5 h
2.2	Transfer functions and state space	1.0 h
2.3	System identification and frequency response	1.0 h
2.4	PID and basic control structures	1.5 h
2.5	Stability and performance analysis	1.0 h

Bring along from today:
the two DC-motor equations, the time constants τ_e and τ_m, friction, and the mass–spring–damper with ω_n and ζ. Tomorrow we learn a tool (the Laplace transform) that turns these ODEs into algebra – and then we control them.

Figure credits

Slide	File (Wikimedia Commons)	Author	License
Brushed DC motor	Motor disassembeled.JPG	Jfer91	CC BY-SA 3.0
Brushless motors	Floppy drive spindle motor open.jpg	Sebastian Koppehel	CC BY 3.0
Induction motor	Wirnik by Zureks.jpg	Zureks	CC BY-SA 3.0
Stepper motor	Positioner Stepper Motor parts.jpg	tony_duell	CC BY 2.0
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Ball screw	BallScrewNut-showing-balls-recirculating-tubes.jpg	Glenn McKechnie (Graibeard)	CC BY-SA 3.0
Gearboxes	Epicyclic gear ratios.png	Wapcaplet	CC BY-SA 3.0
Gearboxes	Harmonic Drive AG strain wave gear set.jpg	Pieceofmetalwork	CC BY-SA 4.0

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